LOGISTIC REGRESSION

Why Convert the Score into a Probability?

The linear model gives a raw score z. For classification, we need a value that behaves like a probability.

01 SCORE

Compute
$z = w^T x + b$

02 SQUASH

Apply sigmoid
$\sigma(z)$

03 PROBABILITY

Get p between
0 and 1

04 LOSS

Compare p with
the true label

x → z → sigmoid → probability p → cross-entropy loss

What Does Probability p Mean?

Probability is the model's estimated likelihood that the input belongs to a class.

BINARY CLASSIFICATION

$p = P(y = 1 \mid x)$

Example: p = 0.80 means the model assigns 80% probability to class 1.

THE OTHER CLASS

$P(y = 0 \mid x) = 1 - p$

If p = 0.80, then class 0 gets 0.20.

Important: probability is a continuous score between 0 and 1. It is not yet the final class label.

NEXT → a threshold converts probability into a class decision

From Raw Score z to Probability p

Example: suppose the model produces z = 2. We convert it using the sigmoid function.

STEP 1 — RAW SCORE

$z = w^T x + b$

Example: $z = 2.00$

STEP 2 — SIGMOID

$p = 1 / (1 + e^{(-z)})$

$p = 1 / (1 + e^{(-2)}) \approx 0.881$

The sigmoid forces any real-valued score into the interval 0 to 1.

NEXT → interpret p as confidence for class 1

What Is a Threshold?

A threshold is the cutoff used to convert a predicted probability into a final class.

PROBABILITY

Model output:
p = 0.72

THRESHOLD

Choose cutoff:
t = 0.50

DECISION

Because 0.72 ≥ 0.50,
predict class 1

Decision rule: if p ≥ threshold → class 1; otherwise → class 0.

Probability tells us confidence. The threshold tells us where to draw the decision boundary.

Does the Threshold Have to Be 50%?

No. A threshold of 0.50 is common, but it is a design choice—not a law.

WHY 0.50 IS COMMON

For logistic regression, $p \geq 0.50$ naturally separates the two classes when the costs are treated similarly.

LOWER THRESHOLD

Example: $t = 0.30$
More samples become positive.
Usually ↑ recall, but also ↑ false positives.

HIGHER THRESHOLD

Example: $t = 0.80$
Require stronger confidence.
Usually ↓ false positives, but may miss positives.

HOW DO WE CHOOSE IT?

Tune the threshold on validation data based on the goal: recall, precision, F1, cost, safety, or another metric.

THRESHOLD EXAMPLES

Same Probability, Different Decisions

Changing the threshold changes the final class even though the model probability stays the same.

EXAMPLE A

$p = 0.40,\ t = 0.50$
$0.40 < 0.50$
→ class 0

EXAMPLE B

$p = 0.40,\ t = 0.30$
$0.40 \geq 0.30$
→ class 1

EXAMPLE C

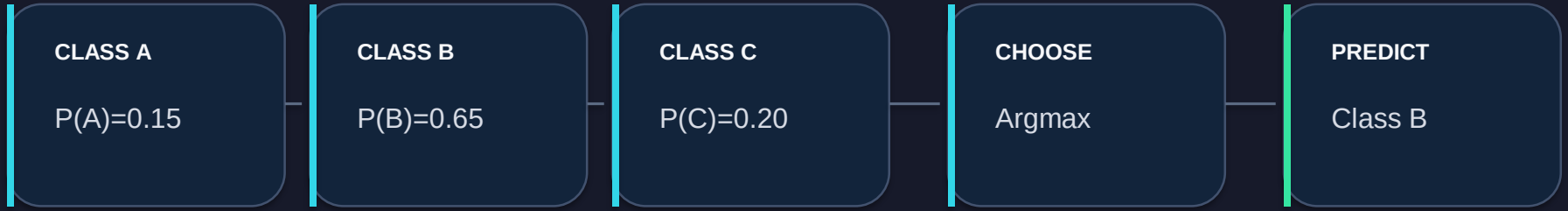
$p = 0.75,\ t = 0.80$
$0.75 < 0.80$
→ class 0

The model did not change. Only the decision rule changed.

Lower threshold = easier to call something positive. Higher threshold = harder to call it positive.

What If We Have More Than Two Classes?

With multiple classes, the model usually produces one probability for each class.



KEY IDEA

Often we choose the class with the highest probability. A separate confidence threshold can still be added for “uncertain / reject” behavior.

What Does Cross-Entropy Measure?

It measures how much probability the model gave to the correct answer.

TRUE LABEL

$y = 1$
The correct class is 1.

MODEL PROBABILITY

$p = 0.881$
The model gives 88.1% to class 1.

LOSS

$L = -\log(0.881)$
$L \approx 0.127$
Small loss = good

WHY THE LOG?

The log makes confident wrong predictions very expensive. If $y = 1$ but $p = 0.10$, loss ≈ 2.303.

Cross-Entropy: Three Step-by-Step Examples

Assume the true label is $y = 1$. Then binary cross-entropy simplifies to $L = -\log(p)$.

GOOD

$p = 0.90$
$L = -\log(0.90)$
≈ 0.105

UNCERTAIN

$p = 0.50$
$L = -\log(0.50)$
≈ 0.693

BAD

$p = 0.10$
$L = -\log(0.10)$
≈ 2.303

PATTERN

As the probability assigned to the correct class decreases, the loss increases sharply.

Training vs. Final Classification

Cross-entropy trains the probability model. The threshold is used afterward to convert probability into a class.

INPUT

Features x

SCORE

$z = w^T x + b$

PROBABILITY

$p = \sigma(z)$

TRAIN

Cross-entropy

DECIDE

Compare p to t

KEY IDEA

Threshold is usually not part of the cross-entropy formula.
Training improves p; thresholding turns p into class 0 or 1.